

14. human CARMIL1 REGION "Necessary for localization at the cell membrane" (35 aa) and mouse Q6EDY6

TOOLS: kmerseek hp_thomas_dill2 k19 (chosen for REGION in Stage 0) and hp_kyte_doolittle2 k19; phmmmer, MMseqs2, MMseqs2 iterative, Foldseek, ProstT5, Reseek; Kyte-Doolittle scan

- kmerseek: call lands on the feature (>= 80% of call inside, >= 30% covered)

call mostly inside the feature, lower IoU than kmerseek

call overlaps the feature but < 50% of the call is inside it

call reaches kmerseek's IoU or higher

no call of this feature type overlaps the feature

the tool's best call is on another target protein

Kyte-Doolittle scan: no target protein, it searches nothing

the Swiss-Prot feature being scored (human) / transferred from (target)

other Swiss-Prot features, labelled by type
- human CARMIL1 (Q5VZK9)

kmerseek hp_thomas_dill2 k19, chosen for REGION (IoU 0.83)

kmerseek hp_kyte_doolittle2 k19, best other alphabet (IoU 0.92)

phmmmer (IoU 0.11)

MMseqs2 (IoU 0.02)

MMseqs2 iterative (IoU 0.02)

Foldseek (IoU 0.09)

ProstT5 (IoU 0.15)

Reseek (IoU 0.65)

Kyte-Doolittle scan

no call

residue position on human CARMIL1 (Q5VZK9) (aa; protein is 1371 aa, window shown)
- mouse Q6EDY6

kmerseek hp_thomas_dill2 k19, chosen for REGION

kmerseek hp_kyte_doolittle2 k19, best other alphabet

phmmmer

MMseqs2

MMseqs2 iterative

Foldseek

ProstT5

Reseek

Kyte-Doolittle scan

its best call is on another target protein

no target: the scan reads the human sequence only

residue position on mouse Q6EDY6 (aa; protein is 1374 aa, window shown)
- Hypothesis: kmerseek places the human REGION feature "Necessary for localization at the cell membrane" on its mouse counterpart and the structure tools do not.
Conclusion: kmerseek IoU 0.83, 24 of 32 residues identical; Foldseek spills; ProstT5 spills; Reseek inside, lower IoU; phmmmer spills; MMseqs2 spills; MMseqs2 iterative spills; Kyte-Doolittle scan did not land; pLDDT 32, disorder fraction 1.0.